

CHRISTIAN LI, KENNY WANG, KEVIN YAO, MATIN SOBHANI, NICK LAFREDO, PARSA REZVANI, RYAN YU

## PURPOSE

Remote wildfire areas often lack real-time monitoring.

## SOLUTION

Drone nodes detect fire/smoke, send GPS telemetry, and help estimate fire location on a live ground-station map.

## OBJECTIVES

- Detect fire/smoke from aerial video
- Transmit drone GPS, heading, and alert status
- Locate fire using 2+ drone observations
- Display alerts on a live map

## TARGET USERS

- Wildfire operations coordinators
- Fire-response and forestry field crews
- Environmental researchers
- Future engineering teams

## SOFTWARE

### Ground Station Interface

- Displays drone positions, alerts, and estimated fire markers

### Vision Pipeline

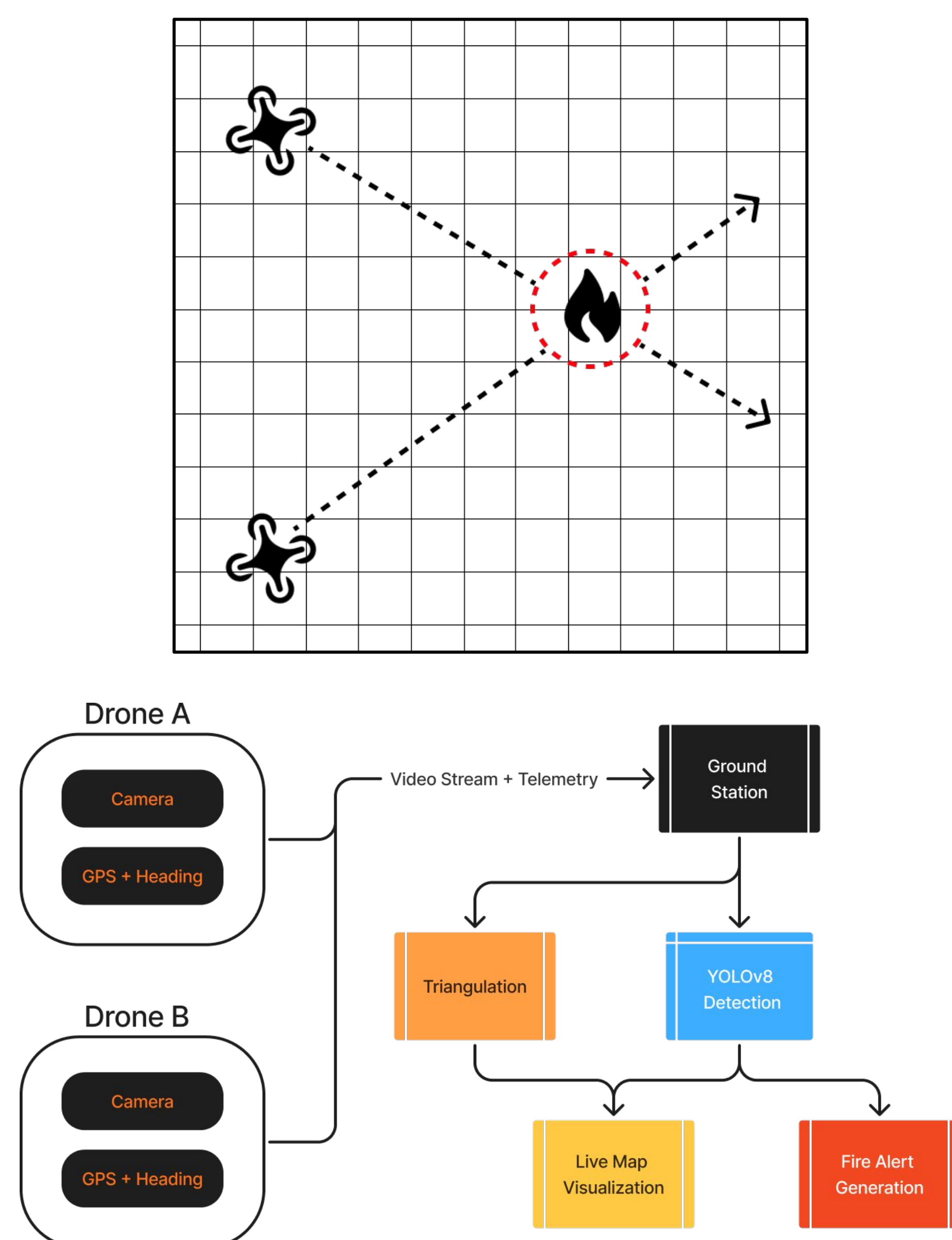
- YOLOv8 fire/smoke detection

### Localization

- Estimates GPS coordinates of detected fire events

### Telemetry Backend

- Parses real-time ESP32 telemetry messages
- Updates active drone and fire-event state



## HARDWARE

### Drone Node

- ESP32 flight/telemetry microcontroller
- ESP32-CAM video module
- GPS receiver for position data

### Wireless Communication

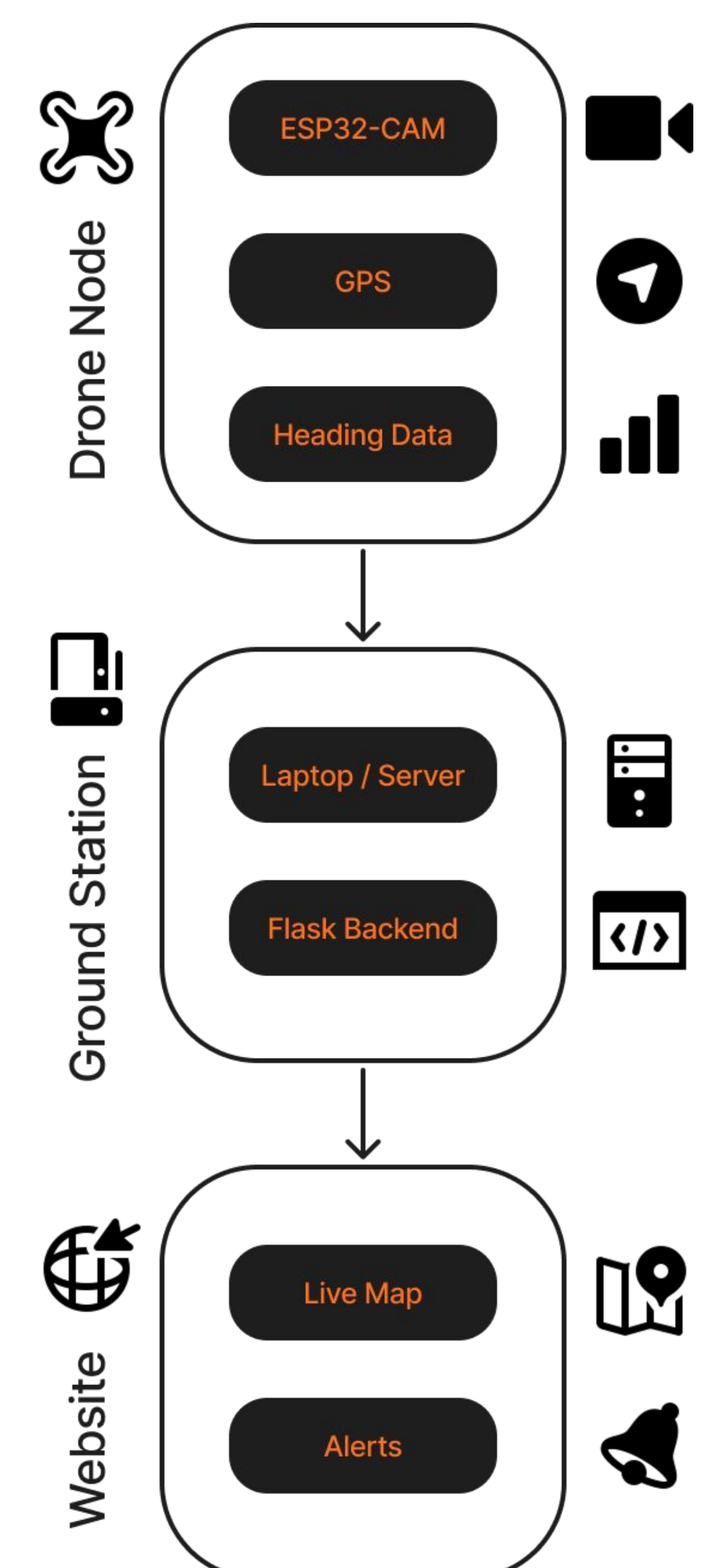
- ESP-NOW broadcast telemetry between drone nodes
- Wi-Fi connection to ground station

### Ground Station

- Receives telemetry and fire alerts
- Displays live map and detection results

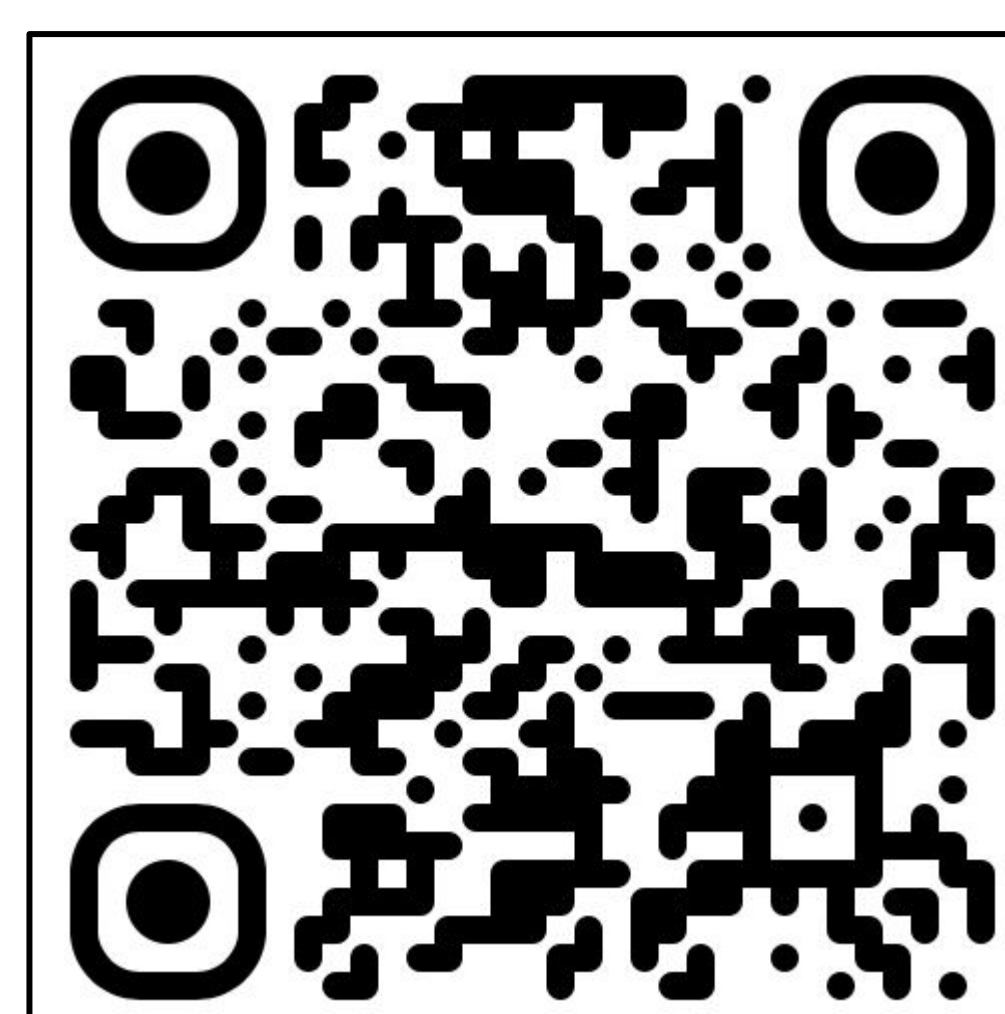
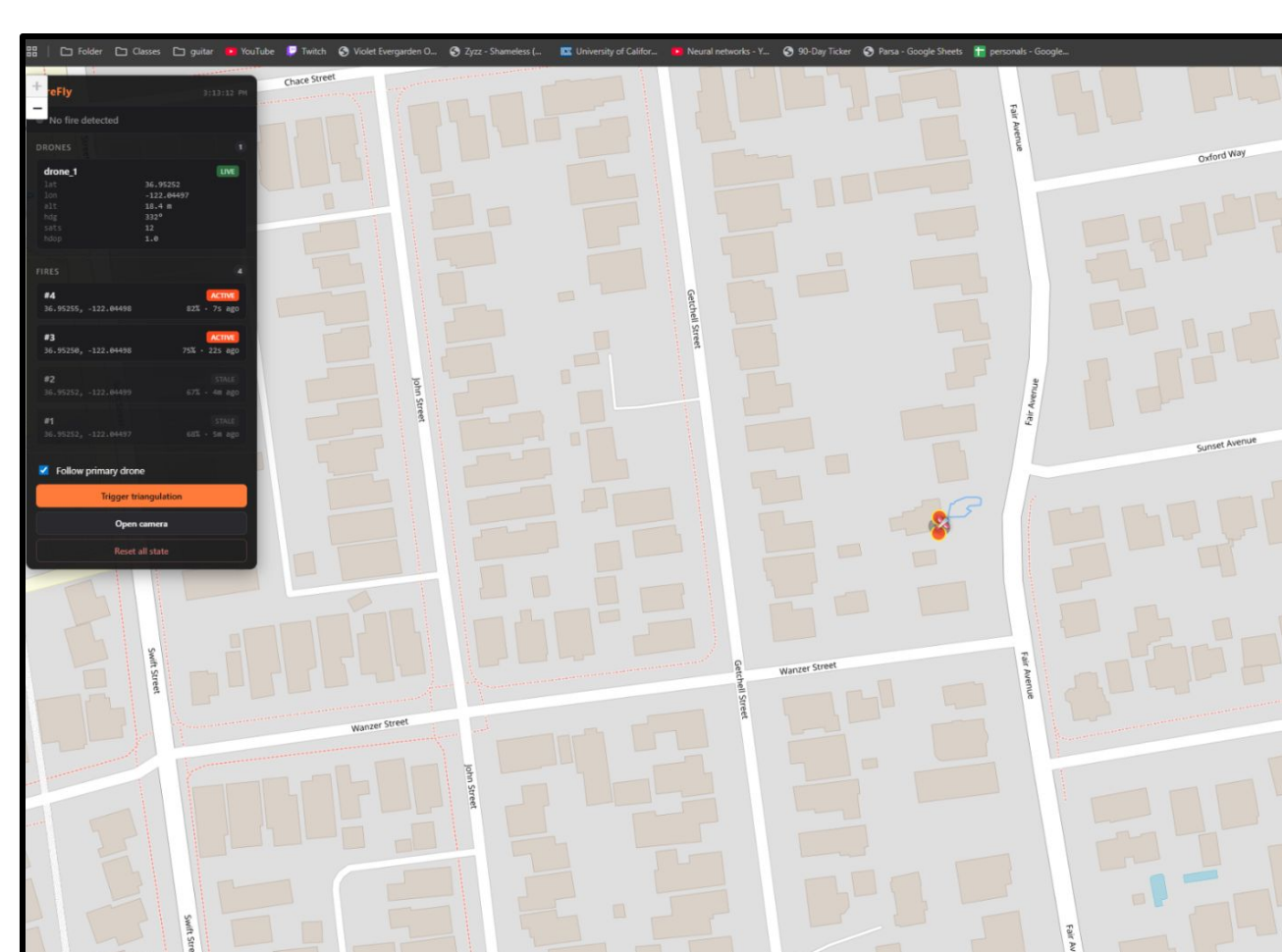
### Sensor Data Collected

- Camera stream for fire/smoke detection
- GPS coordinates

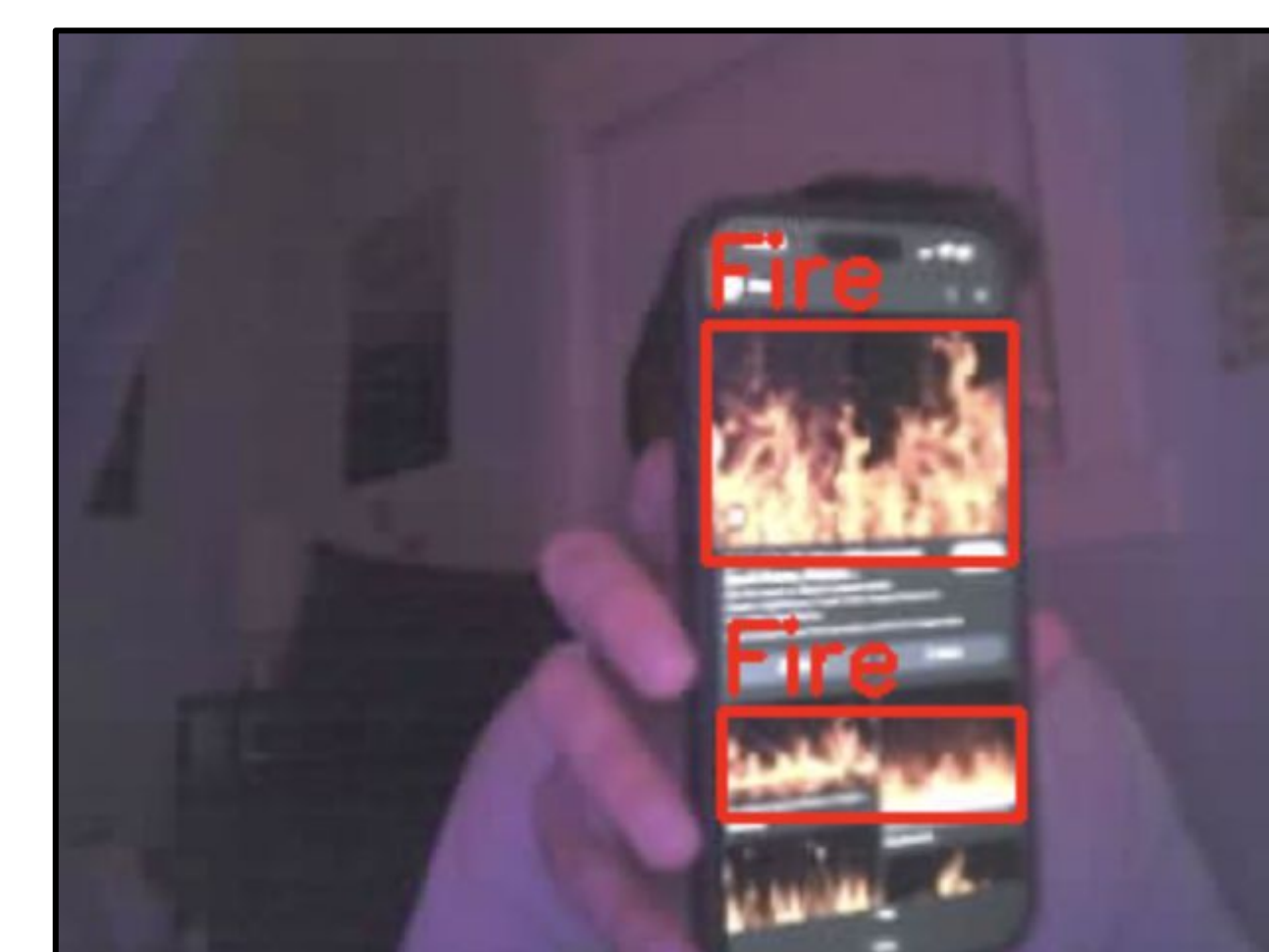


## DEMO AND VALIDATION

- Ground Station Map
- Drone telemetry and estimated fire coordinates are displayed in real time



msobhani@ucsc.edu



- Fire Detection View
- YOLOv8 identifies sustained fire and smoke from the camera stream